

Weather radar mode on the ND

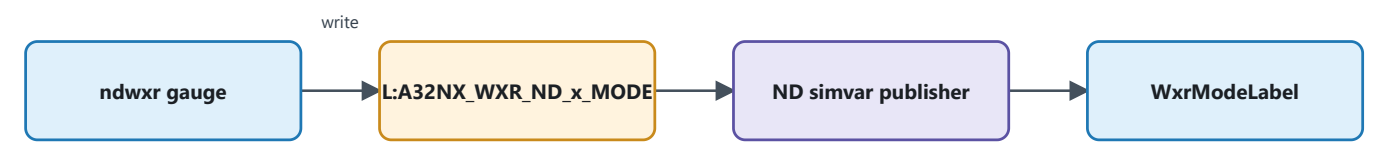
WX, WX+T, TURB, MAP or WXR OFF shown on the map pages

Branch	feature/nd-wxr-mode-label
Aircraft	A32NX and A380X
Commits	a5ecd08cf (PR 3)
Base	feature/nd-terrain
Status	In master. PR draft 03.

Specification

- The gauge publishes the mode as `L:A32NX_WXR_ND_{L,R}_MODE` (0 none, 1 WX, 2 WX+T, 3 TURB, 4 MAP, 5 WXR OFF), only while the radar is selected on a map page and not replaced by the terrain.
- The ND draws it with the new `WxrModeLabel1` (green, right of the page; WXR OFF white on the A32NX, A320 FCOM DSC-34-SURV-30-30). The A380X shows its own list: WX / MAP / WXR OFF (A380 FCOM DSC-34-20-30-20).

■ Component added or changed by the feature ■ MSFS / simulator ■ Aircraft system (systems host, FMS, FWS) ■ Display / page
■ Data, LVar, storage ■ Crew, flypad, external program



Architecture

PR description (from the PR draft)

From `pr-drafts/03-wx-mode-label.md`, section "Summary of Changes" onwards.

Summary of Changes

Nothing on the ND told the crew which radar mode was selected. The mode (WX, WX+T, TURB or MAP) is now shown in green on the right of the ARC and ROSE NAV pages while the radar is on, on both aircraft.

- The native radar gauge already knows whether the radar is selected and shown (the WX SYS knob on the A32NX; the WX overlay, the WXR & TAWS system and the SURV CONTROLS buttons on the A380X), so it publishes the text as a number in `A32NX_WXR_ND_{L,R}_MODE` (0 none, 1 WX, 2 WX+T, 3 TURB, 4 MAP), only while the radar is selected on an ARC or ROSE NAV page and not while the terrain takes its place.
- The ND reads it through its simvar publisher and the new `WxrModeLabel` component draws the text, so the gating rules are not duplicated in the ND code.
- The A380X has no TURB-only mode: its SURV CONTROLS page gives WX, WX+T and MAP.
- The text is also shown on the ground, where the radar itself does not draw.

Cockpit API Changes

New LVars, written by the `ndwxr` gauge and read by the ND. Already documented in `fbw-a32nx/docs/a320-simvars.md` (new "Weather Radar (ATA 34)" section) and `fbw-a380x/docs/a380-simvars.md` (new "Surveillance ATA 34" section), in the commit of this PR:

Var	Type	Description
<code>L:A32NX_WXR_ND_L_MODE</code>	Enum	Weather radar mode text of the left (CPT) ND: 0 none, 1 WX, 2 WX+T, 3 TURB, 4 MAP
<code>L:A32NX_WXR_ND_R_MODE</code>	Enum	Same for the right (F/O) ND

Screenshots (if necessary)

TO ADD: ARC and ROSE NAV with the label in each mode (WX, WX+T, TURB, MAP), and without the radar (no label). Before: same page, no label.

References

TO ADD: real ND photo/FCOM screenshot showing the weather radar mode text (position and colour) on the A320 and the A380 ND.

Additional context

Position: x 744, y 590 of the ND svg, `Green FontSmall1`, right-aligned, after `TerrainMapThresholds`. The labels are not localised (like the other ND texts).

Discord username (if different from GitHub): **TO ADD**

Testing instructions

1. A32NX: WX SYS on SYS 1, ND in ARC then ROSE NAV: the mode text follows the MODE knob (WX, WX+T, TURB, MAP). WX SYS OFF: no text.
2. Other ND modes (PLAN, ROSE ILS, ROSE VOR): no text.
3. Turn TERR ON ND on (with the terrain PR): the text goes while the terrain is drawn.
4. A380X: WX overlay + WXR & TAWS system selected: text follows the SURV CONTROLS TURB / MODE buttons (WX, WX+T, MAP). Range index 0 (OANS): none.
5. On the ground the text is shown although the radar does not draw.